$\mathbf{T^3}$ as a Closed Information-Processing Environment

Mathematical Requirements for Stable Discrete Information
in a Compact Flat Manifold

Vladimer Merebashvili

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Abstract

We derive the minimal mathematical structures required for the three-dimensional flat torus $T^3 = \mathbb{R}^3/L\mathbb{Z}^3$ to function as a closed, stable, discrete information-processing environment. The question is not what external physics the geometry implies, but what the geometry itself *requires*: what conditions must hold for information to be stably encoded, reliably clocked, conserved under flow, and filtered against noise within a compact flat manifold with periodic boundary conditions. We identify five minimal requirements — stability, a well-defined clock, conservation, laminarity, and noise rejection — and show that each is satisfied by a unique mathematical object that T^3 provides automatically, without external design. The stable information vocabulary consists of exactly $3^3 - 1 = 26$ discrete vortex states in three symmetry families, determined by the first Betti number $b_1(T^3) = 3$. The natural clock is the lowest cavity resonance $f_1 = c/L$. Information is conserved by seven independent quantities: four continuous Noether charges and three discrete topological winding numbers. Laminar flow is the unique consequence of the Laplace equation on a compact flat manifold. Noise rejection operates at two complementary levels: the Landau energetic filter decays high-winding states within one clock period, while Legendre's three-square theorem geometrically forbids states at shells $|\mathbf{n}|^2 \in \{7, 15, 23, \ldots\}$. We show that these five mechanisms are not only each satisfied but jointly sufficient for closure, and that the structure is independent of the four external scale parameters (L, c, ρ_s, ξ). The closed T^3 information system is a topological fact.

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1 Introduction and Motivation

1.1 The Problem of Closed Information

A closed information-processing system is one that encodes, transmits, transforms, and stores information entirely within its own boundary, without leakage to or dependence on an external sink or source. Classical information theory, following Shannon [1], asks how efficiently a channel can transmit information subject to noise, and answers with coding theorems that describe the capacity of an *open* channel between a sender and a receiver. The present paper asks a different question: what geometry is *minimally sufficient* to support information processing without any channel at all — as a closed, self-contained system?

This is not a question about efficiency or capacity in the Shannon sense. It is a question about structural necessity. We do not ask how much information T^3 can transmit, but whether it can guarantee that information encoded in it is stable against perturbation, reliably timed, conserved under evolution, protected against distortion, and self-correcting against noise — purely by virtue of its topology. The answer, which we derive in detail, is yes: and the specific mathematical objects that provide each guarantee are uniquely determined by the topology.

The topology-based approach has a fundamental advantage over coding-based approaches: it provides a *structural* guarantee rather than a probabilistic one. A Shannon capacity theorem says that arbitrarily reliable transmission is possible given sufficiently long codes, but it does not eliminate the noise floor — it manages it. Topology eliminates it: a winding number is either unchanged (zero perturbation) or changed (a discrete event requiring finite energy). There is no analog of a soft error that partially corrupts a winding number. The topological gap between integer winding states is absolute.

1.2 Why T^3?

This paper takes the three-dimensional flat torus $T^3 = \mathbb{R}^3/L\mathbb{Z}^3$ as its starting point. The choice is not arbitrary: T^3 is the smallest compact flat manifold that satisfies all four irreducible geometric requirements for a closed information environment.

A closed information environment must be *finite*: signals cannot escape to infinity, so the manifold must have finite volume. It must have *no boundary*: edges are sites of information loss, reflection, or leakage, so the manifold must be boundaryless. It must be *flat*: curvature focuses and defocuses signals differentially, creating path-length differences and potential interference, so zero Gaussian curvature $K = 0$ is required for distortion-free propagation. And it must carry *multiple independent topological channels*: a single winding direction gives only one bit of topological addressing, which is insufficient for a rich vocabulary; the system needs at least three independent channels to produce the $3^3 - 1 = 26$-state vocabulary we will derive.

The three-torus satisfies all four conditions exactly: finite volume $V = L^3$, no boundary $\partial T^3 = \varnothing$, everywhere zero curvature $K = 0$, and first Betti number $b_1(T^3) = 3$ providing exactly three independent topological channels.

To see why T^3 is *minimal*, consider the alternatives. The circle T^1 has $b_1 = 1$: one channel, yielding at most $3^1 - 1 = 2$ stable states under the stability criterion we derive. This is too impoverished for interesting information processing. The two-torus T^2 has $b_1 = 2$: two channels, giving $3^2 - 1 = 8$ stable states, which is richer but lacks the third channel needed to produce the three-family structure whose non-isomorphism forces a mandatory coupling hierarchy. Higher tori T^n for $n > 3$ give larger vocabularies — $3^n - 1$ states — but are not minimal: T^3 is the threshold at which the vocabulary first acquires the full three-family structure with its structural consequences. We derive all of these results from the geometry; here we simply note that T^3 sits at the minimum dimensionality for the structure to be complete.

1.3 Structure of the Paper

We identify five minimal requirements for a closed information environment, derive from T^3 topology alone that all five are automatically and uniquely satisfied, and then show that the five together are jointly sufficient for closure. The requirements are:

1. **Stability**: information states must be discrete and stable. T^3 with a $U(1)$ order parameter produces exactly $3^3 - 1 = 26$ stable vortex states (Section 3).

2. **Clock**: the system needs an intrinsic timing reference. The cavity resonance spectrum provides a canonical clock at $f_1 = c/L$ (Section 4).

3. **Conservation**: information must not be destroyed. Seven independent conservation laws guarantee this (Section 5).

4. **Laminarity**: information must flow without distortion. The Laplace equation forces uniquely laminar flow (Section 6).

5. **Noise rejection**: the system must self-restore. The Landau and Legendre filters together provide a two-layer noise floor (Section 7).

Section 2 establishes the T^3 topology. Sections 3–7 derive each requirement. Section 8 assembles the complete system and establishes its sufficiency for closure. Section 9 identifies the genuine boundaries of the framework. The paper closes with a summary table and conclusions.

Computational resources. The complete Python implementation of the TVL classification map — including 44 self-test assertions, a closed-form stability proof, and computational verification of the $B_3 \cong \mathfrak{so}(7)$ root system — is available as open-source code at `github.com/merebashvili/topological-vortex-logic`. A companion technical report providing full proofs and worked examples for all five theorems is included as ancillary material.

2 The T^3 Manifold: Topology and Fundamental Properties

2.1 Definition and Construction

The three-torus is defined as the quotient

$$T^3 = \mathbb{R}^3/L\mathbb{Z}^3,$$

where $L > 0$ is the periodicity scale. Concretely, T^3 is the cube $[0, L)^3$ with opposite faces identified: any point $\mathbf{x}$ is identified with $\mathbf{x} + L\mathbf{n}$ for all $\mathbf{n} \in \mathbb{Z}^3$. This produces a compact 3-manifold with finite volume $V = L^3$, no boundary, and no preferred interior point. The manifold inherits zero Gaussian curvature from $\mathbb{R}^3$: the identifications are isometries, not curvature-inducing deformations.

2.2 Topological Invariants and Their Roles

T^3 carries fixed topological invariants that constrain every structure derived from it:

Invariant	Value	Role
Euler characteristic $\chi(T^3)$	0	Non-vanishing vector fields exist; flow has no forced zeros
First Betti number b_1	3	Three independent topological channels
Fundamental group $\pi_1(T^3)$	$\mathbb{Z}^3$	Three conserved integer winding charges
Homology $H_1(T^3, \mathbb{Z})$	$\mathbb{Z}^3$	Three independent topological invariants
de Rham H^1	$\mathbb{R}^3$	Three independent conservation currents
Spin structures	$8 = 2^3$	Eight boundary condition choices for spinor fields

Two constraints deserve special attention. The Euler characteristic $\chi = 0$ matters because by the Poincaré-Hopf theorem, $\chi = 0$ guarantees the existence of a smooth non-vanishing vector field on T^3. This means a flow field can be defined everywhere without forced zeros, which would otherwise act as vortex-free regions disrupting the information encoding. The Betti number $b_1 = 3$ provides exactly three independent topological channels — the substrate for the three-component winding address.

2.3 The Z_3 Orbifold and Its Effect on the Winding Lattice

The structure derived in this paper goes beyond the bare torus T^3: we equip it with a $\mathbb{Z}_3$ orbifold symmetry, specifically $\mathbb{Z}_3 = \text{center}(\text{SU}(3))$. This orbifold acts on the winding lattice $\mathbb{Z}^3$ via the holonomy:

$$V = \text{diag}(\omega, \omega, \omega), \qquad \omega = e^{2\pi i/3}. \tag{1}$$

A vortex state $\mathbf{w}$ acquires phase ω^{q_3} under V, where $q_3 = (w_1 + w_2 + w_3) \bmod 3$ is the $\mathbb{Z}_3$ gauge charge.

The significance of this orbifold is that it partitions the 26-state vocabulary into three charge sectors without breaking any of the stability properties. It adds a *charge label* to each state that is preserved by all smooth field evolutions — a topological invariant derived purely from the holonomy structure. The $\mathbb{Z}_3 = \text{center}(\text{SU}(3))$ choice is the natural symmetry for a system in which the three winding directions play symmetric roles: the cyclic group $\mathbb{Z}_3$ permutes the three coordinates, and its center action provides the cleanest charge structure. The specific charge partition — eight zero-charge states, nine unit-charge states, nine anti-unit-charge states — emerges from the geometry without any external input.

3 Requirement 1: Stable Discrete Information States

3.1 Why Topology and Not Coding

The standard approach to stable information storage uses redundancy: encode each bit multiple times and use majority voting to correct errors. This approach works statistically but requires external resources (extra space, extra energy for error-correction operations) and provides probabilistic rather than absolute guarantees.

Topology provides a qualitatively different mechanism. A topological invariant is an integer that *cannot* change under any continuous deformation of the field configuration. It is not unlikely to change — it is impossible without a discrete, finite-energy event (a singularity passing through the field). There is no analogy to a soft error that partially shifts a topological charge. The stability is absolute within the stable energy regime.

On T^3, the relevant topological invariants are winding numbers around the three generating circles. A field configuration $\psi : T^3 \to U(1)$ that winds w_i times around the i-th circle carries the integer $w_i = (1/2\pi) \oint_{C_i} d\theta$ as a conserved label. The target space $U(1)$ is chosen because $\pi_1(U(1)) = \mathbb{Z}$, so each of the three circles in T^3 contributes one independent integer winding number, matching $\pi_1(T^3) = \mathbb{Z}^3$.

3.2 The Energy Scaling and Landau Instability

For winding numbers to be information labels, they must be energetically stable. In a superfluid — the natural physical realisation of a $U(1)$ order parameter — the velocity field near a vortex with winding n scales as $v \propto n/r$, where r is the distance from the vortex core. Integrating the kinetic energy density $\frac{1}{2}\rho_s v^2 \propto n^2/r^2$ over the manifold (infrared cutoff L, ultraviolet cutoff ξ) gives

$$E(w_i = n) = n^2 \varepsilon_0, \qquad \varepsilon_0 = \frac{\hbar \rho_s}{m} \pi \ln\left(\frac{L}{\xi}\right), \tag{2}$$

where $\hbar$ is the reduced Planck constant, ρ_s is the superfluid density, m is the particle mass, and ξ is the vortex core radius (coherence length). The n^2 scaling is the Landau stability criterion [2]: a doubly-wound vortex ($n = 2$) has energy $4\varepsilon_0$, while two separate unit-wound vortices have combined energy $2\varepsilon_0$. Since $4\varepsilon_0 > 2\varepsilon_0$, the doubly-wound state spontaneously decays. The information consequence is that only unit winding ($|n| = 1$) is stable along any single axis.

For the three-component winding vector, the energy generalises to $E(\mathbf{w}) = |\mathbf{w}|^2 \varepsilon_0$ [3], and stability requires that no split into two non-vacuum states lowers the energy:

Theorem 1 (Stability)**.** *A vortex state $\mathbf{w} \in \mathbb{Z}^3$ is energetically stable against spontaneous splitting if and only if $|\mathbf{w}|^2 \leq 3$.*

Proof. A split $\mathbf{w} \to \mathbf{u} + (\mathbf{w} - \mathbf{u})$ is favourable iff $\mathbf{w} \cdot \mathbf{u} > |\mathbf{u}|^2$.

Unstable direction. In $\mathbb{Z}^3$, the maximum squared norm with all components in $\{-1, 0, 1\}$ is 3. Therefore any $\mathbf{w}$ with $|\mathbf{w}|^2 \geq 4$ must have at least one component $|w_i| \geq 2$. Choosing

$\mathbf{u} = \mathrm{sign}(w_i)\,\mathbf{e}_i$ gives $\mathbf{w} \cdot \mathbf{u} = |w_i| \geq 2 > 1 = |\mathbf{u}|^2$. The split is favourable for *all* $|\mathbf{w}|^2 \geq 4$ by a single algebraic argument.

Stable direction. Exhaustive computation over all $\mathbf{w}$ with $|\mathbf{w}|^2 \leq 3$ and all valid splits confirms $E_{\text{split}} \geq E_{\text{original}}$ in every case. The marginal case is $(1, 1, 1)$ with three neutral splits ($E_{\text{split}} = 3 = E_{\text{original}}$). □

3.3 The 26-State Vocabulary

The stable vocabulary is

$$\Omega_{\text{stable}} = \left\{\mathbf{w} \in \{-1, 0, 1\}^3 \setminus \{\mathbf{0}\}\right\}, \qquad |\Omega_{\text{stable}}| = 3^3 - 1 = 26. \tag{3}$$

The $3^3 - 1$ formula reflects a clean combinatorial fact: the stable states are exactly the non-zero elements of the set $\{-1, 0, 1\}^3$, which has $3^3 = 27$ elements in total, minus the vacuum $\mathbf{0}$.

The 26 states organise into three families by $|\mathbf{w}|^2$:

Family	$\lvert\mathbf{w}\rvert^2$	Count	Active channels	Example
Face	1	6	One	$(\pm1, 0, 0)$ and permutations
Edge	2	12	Two	$(\pm1, \pm1, 0)$ and permutations
Corner	3	8	Three	$(\pm1, \pm1, \pm1)$
Total		26	$3^3 - 1$	

The three-family structure is forced: $b_1(T^3) = 3$ independent winding directions, Landau stability restricts components to $\{-1, 0, 1\}$, and the number of simultaneously active channels ranges from one to three. A fourth family would need $|\mathbf{w}|^2 = 4$, which is forbidden by Theorem 1.

3.4 The Non-Isomorphism Theorem

Theorem 2 (Non-Isomorphism)**.** *The three vortex families are pairwise non-isomorphic as $\mathbb{Z}_3$-modules under the cyclic permutation $g : (w_1, w_2, w_3) \mapsto (w_2, w_3, w_1)$.*

Proof. A $\mathbb{Z}_3$-isomorphism maps g-fixed points to g-fixed points bijectively. A vector $\mathbf{w}$ is fixed by g iff $w_1 = w_2 = w_3$.

- Face states have exactly one non-zero component: impossible to have $w_1 = w_2 = w_3$ unless all zero. Fixed-point count: 0.
- Edge states have exactly two non-zero components: same argument. Fixed-point count: 0.
- Corner states have all components ±1: $w_1 = w_2 = w_3$ gives $(1, 1, 1)$ and $(-1, -1, -1)$. Fixed-point count: 2.

Since Face and Edge both have count 0 while Corner has count 2, Face $\not\cong$ Corner and Edge $\not\cong$ Corner.

For Face vs. Edge: both have zero fixed points, so we use the ρ_0 multiplicity (dimension of the trivial-representation subspace). The character formula gives Face = 2, Edge = 4. Since $2 \neq 4$, Face $\not\cong$ Edge. No two families are isomorphic. □

The proof establishes two distinct pairs of module-theoretic invariants. The fixed-point count distinguishes Corner from the others; the ρ_0 multiplicity distinguishes Face from Edge. Neither invariant alone suffices for all three separations; both are needed. This is not a deficiency — it reflects a genuine algebraic richness in the three-family structure.

Corollary. Any coupling that discriminates families by their $\mathbb{Z}_3$-module structure must assign distinct coupling strengths to the three families. The ordering is fixed by the module invariants; only the overall energy scale remains free.

The significance of this corollary extends beyond information theory. It says that any physical interaction sensitive to the winding structure — a Higgs-like coupling, for instance — is *forced* by the geometry to treat the three families unequally. This is a mandatory hierarchy with no free ordering parameter: the geometry determines which family couples most strongly and which least.

3.5 The Symmetry Algebra of the Vocabulary

The full 26-state vocabulary is invariant under the *cubic symmetry group* O_h: the group of signed permutations of the three winding components, of order 48. This is the group of all linear maps $\mathbb{Z}^3 \to \mathbb{Z}^3$ that preserve $\{-1, 0, 1\}^3 \setminus \{\mathbf{0}\}$. Every such map sends face states to face states, edge states to edge states, and corner states to corner states (within each family, all states are related by O_h symmetry).

The 18 face-and-edge states form the root system $B_3 \cong \mathfrak{so}(7)$, verified by checking the four root system axioms (Appendix A). The Cartan matrix of the simple roots $\alpha_1 = (1, -1, 0)$, $\alpha_2 = (0, 1, -1)$, $\alpha_3 = (0, 0, 1)$ is the standard B_3 matrix $\mathrm{diag}(2, 2, 2) +$ off-diagonal entries. The Weyl group of B_3 has order $3! \times 2^3 = 48 = |O_h|$, confirming that the symmetry group of the vocabulary is precisely the Weyl group of the underlying root system.

The information-theoretic significance is that O_h is not an externally imposed structure: it is the *natural symmetry of the addressing system.* Any two states related by an O_h transformation are equivalent modulo the orientation of the three winding directions. The vocabulary is as symmetric as it could be subject to the constraint of three families.

3.6 The $\mathbb{Z}_3$ Charge as a Geometric Selector

The $\mathbb{Z}_3$ orbifold holonomy assigns charge $q_3 = (w_1 + w_2 + w_3) \bmod 3$ to each state. This charge partitions the 26 states:

$$q_3 = 0 : 8 \text{ states (singlets)}, \quad q_3 = 1 : 9 \text{ states (unit charge)}, \quad q_3 = 2 : 9 \text{ states (anti-unit charge)}.$$

The charge $q_3 = 0$ selects the states that are "transparent" to the $\mathbb{Z}_3$ holonomy: the two corner diagonals $(1, 1, 1)$ and $(-1, -1, -1)$, and the six traceless edge states with zero trace. The $q_3 \neq 0$ states are those that transform non-trivially under the holonomy and acquire a phase under the $\mathbb{Z}_3$ symmetry operation.

This geometric separation has a direct consequence for the information system: the $q_3 = 0$ and $q_3 \neq 0$ states cannot be confused by any operation that preserves the holonomy structure. The charge label is a topological invariant — preserved under all continuous deformations — and thus provides an additional dimension of addressing beyond the winding vector itself.

4 Requirement 2: A Well-Defined Clock

4.1 Why an Internal Clock Is Necessary

For a sequence of information transitions to be coherent — for transitions to have a definite ordering and a well-defined rate — the system needs a timing reference. An open system can use an external clock (an oscillator outside the system boundary), but a closed system with no external reference must derive its timing from its own geometry. The timing reference must be intrinsic: derivable from the manifold's properties alone, without material constants or external fields.

The natural intrinsic clock is the *cavity resonance spectrum*: the set of frequencies at which a wave propagating through T^3 returns to its starting phase after one traversal. These frequencies are determined by the allowed wavevectors, which are quantised by the periodic boundary conditions.

4.2 The Cavity Resonance Spectrum

On T^3 with side length L, the allowed wavevectors are $\mathbf{k_n} = (2\pi/L)\,\mathbf{n}$ for $\mathbf{n} \in \mathbb{Z}^3$. The corresponding frequencies are $f_\mathbf{n} = c|\mathbf{n}|/L$, where c is the characteristic signal propagation speed of the medium (the speed of sound in a superfluid, or more generally the wave speed of the physical realisation). Organising by shell $|\mathbf{n}|^2$:

Shell $\lvert\mathbf{n}\rvert^2$	Frequency	Degeneracy	Analog
1	$f_1 = c/L$	6	Face family
2	$f_2 = c\sqrt{2}/L$	12	Edge family
3	$f_3 = c\sqrt{3}/L$	8	Corner family

The fundamental clock is $f_1 = c/L$: the frequency at which a wave traverses one side length and returns in phase [4]. In terms of the equivalent radius $R \equiv L/(2\pi)$: $f_1 = c/(2\pi R)$.

4.3 The Clock-State Correspondence

The relationship between the clock spectrum and the state families is not coincidental: both are indexed by the same integer lattice $\mathbb{Z}^3$. The winding states are labelled by $\mathbf{w} \in \mathbb{Z}^3$ with energy $E = |\mathbf{w}|^2\varepsilon_0$, while the clock modes are labelled by $\mathbf{n} \in \mathbb{Z}^3$ with frequency $f = c|\mathbf{n}|/L$. The three families and the three clock shells are the *same structure seen from two perspectives*: energetic (winding stability) and spectral (cavity resonance).

The clock ratios $f_1 : f_2 : f_3 = 1 : \sqrt{2} : \sqrt{3}$ differ from the energy ratios $E_{\text{face}} : E_{\text{edge}} : E_{\text{corner}} = 1 : 2 : 3$ only because energy goes as $|\mathbf{w}|^2$ while frequency goes as $|\mathbf{n}|$. Both depend on the same norm of the same integer lattice. A state transition within the face family occurs at energy $\Delta E \sim \varepsilon_0$, resonantly matched to a clock period $t_1 = 1/f_1 = L/c$. The clock and the vocabulary are locked to the same underlying structure.

4.4 Clock Harmonics and Discrete-Time Processing

Along any single principal axis, the resonance spectrum provides integer harmonics $f_n = n \cdot f_1$ for all $n \in \mathbb{Z}^+$: processes locked to the same axis are automatically synchronised. The fundamental time quantum is

$$t_q = \frac{1}{f_1} = \frac{L}{c}, \tag{4}$$

the time for a signal to traverse one side of the torus.

An important distinction follows from §4.2. The three shell frequencies $f_1 : f_2 : f_3 = 1 : \sqrt{2} : \sqrt{3}$ are *mutually incommensurate*: $\sqrt{2}$ and $\sqrt{3}$ are irrational, so face, edge, and corner modes do not share a common period and do not automatically synchronise with each other. The role of f_1 is therefore that of a *master clock*: it sets the fundamental time quantum t_q that gates all state transitions. The edge and corner modes (f_2, f_3) function as stable vocabulary states whose characteristic frequencies are geometrically fixed by the lattice, but which are not harmonics of f_1.

Transitions coherent with the master clock occur at integer multiples of t_q; non-coherent transitions correspond to transients damped by the cavity Q-factor. The manifold acts as a discrete-time processor with sampling rate f_1 and Nyquist aliasing protection up to $f_1/2$ along each principal axis.

5 Requirement 3: Information Conservation

5.1 Continuous Laws from Noether's Theorem

T^3 carries four continuous symmetries, each yielding a conserved quantity by Noether's theorem [5]:

Symmetry	Conserved quantity	Information role
Translation x_1	Momentum $p_1 = (h/L)n_1$	Address component 1
Translation x_2	Momentum $p_2 = (h/L)n_2$	Address component 2
Translation x_3	Momentum $p_3 = (h/L)n_3$	Address component 3
Time translation	Energy $E = \|\mathbf{n}\|^2\varepsilon_0$	State stability

The three spatial momenta are quantised at values $p_i = (h/L)n_i$, where h is Planck's constant, because the torus has *discrete* translational symmetry: not invariance under continuous translation, but invariance under translation by integer multiples of L. This quantisation means the momenta take values in a discrete set, which is exactly the condition for them to serve as address components.

5.2 Topological Conservation Laws

Beyond Noether's theorem, T^3 provides three further independent conservation laws from an entirely different source: the fundamental group $\pi_1(T^3) = \mathbb{Z}^3$. The winding numbers (w_1, w_2, w_3) are topological invariants — they cannot change under any smooth evolution of the field. In the stable regime:

$$\frac{dw_i}{dt} = 0 \quad (i = 1, 2, 3) \qquad \text{between transitions.} \tag{5}$$

Each component w_i is individually conserved; this is stronger than mere conservation of the sum $\sum w_i$.

5.3 The Qualitative Difference Between the Two Types

The Noether charges (momenta, energy) are *continuous* quantities: they can take any quantised rational value, and perturbations can shift them by arbitrarily small amounts subject to the quantisation constraint. In an open system, external noise can alter them gradually. The topological charges (winding numbers) are *qualitatively different*: they are integers protected by the topology of the field configuration. A perturbation that alters a winding number must pass the field through a singularity — a discrete, finite-energy event. Small continuous perturbations simply cannot change a winding number, regardless of their amplitude, as long as they remain below the singularity threshold.

For information addressing, this distinction is crucial. An address that can drift continuously is an unreliable address. The topological protection guarantees that the winding address $\mathbf{w} = (w_1, w_2, w_3) \in \{-1, 0, 1\}^3 \setminus \{\mathbf{0}\}$ is absolutely stable against any perturbation that does not cross a singularity. The address is binary-precise: it either stays exactly at its current integer value or, if a singularity occurs, makes a discrete jump to a different integer value. There is no possibility of a "half-corrupted" address.

5.4 Total: Seven Conserved Quantities

Combining the two types gives seven independent conserved quantities: four continuous (energy and three momenta) and three discrete (winding numbers). These exhaust the independent conservation laws of T^3: the de Rham cohomology $H^1(T^3) = \mathbb{R}^3$ provides three conservation

currents, the temporal symmetry provides energy, and $\pi_1(T^3) = \mathbb{Z}^3$ provides the three winding charges. No further independent conservation law is available from the topology alone.

5.5 Transition Events and the Conservation Structure

Between transitions, all seven quantities are conserved. A transition is a discrete event: the creation or annihilation of a vortex-antivortex pair. A pair of states $\mathbf{w}$ and $-\mathbf{w}$ can annihilate to the vacuum $\mathbf{0}$, releasing energy $2|\mathbf{w}|^2\varepsilon_0$. Conversely, energy input from the environment (or from an adjacent state) can create a pair from the vacuum.

The transition rule is constrained by the $\mathbb{Z}^3$ composition law: a transition $\mathbf{w} \to \mathbf{w}'$ must satisfy $\mathbf{w}' = \mathbf{w} + \Delta\mathbf{w}$ where $\Delta\mathbf{w}$ is a stable winding vector (or zero). This follows from winding number conservation: the total winding must be preserved modulo the pair-creation/annihilation events. The vocabulary Ω_{stable} is closed under this transition rule, so information transitions within the vocabulary never produce states outside it.

5.6 Conservation and Error Resistance

The topological conservation has a direct implication for error resistance. Suppose a winding address w_i is subject to a continuous perturbation — environmental noise, thermal fluctuation, or an imprecise read-out. Because winding numbers are integers protected by topology, such a perturbation has *no effect* on the address as long as it remains below the singularity energy threshold ε_0. The perturbation cannot partially shift w_i from 1 to 1.3 and then to 2: it either leaves $w_i = 1$ unchanged, or if it exceeds the threshold, it makes w_i jump discretely to a different integer. The information system thus has a built-in error gap of size ε_0: perturbations smaller than this cannot corrupt an address. This is not probabilistic error correction — it is a structural guarantee.

6 Requirement 4: Laminar Information Flow

6.1 Why Laminarity Is Required

Information flowing through the manifold must not compress or rarefy locally. If the information density (the concentration of occupied states per unit volume) could vary, it would create regions of high density where states are difficult to distinguish, and regions of low density where the effective channel is underused. The condition that prevents this is divergence-free flow: $\nabla\cdot\mathbf{v} = 0$. Whatever information enters any region exits at the same rate; no net accumulation or depletion occurs.

6.2 The Laplace Equation and Its Consequences

The simplest divergence-free flow is a potential flow $\mathbf{v} = \nabla\Phi$, where Φ is a scalar potential. Combining with the incompressibility condition gives $\nabla^2\Phi = 0$ — the Laplace equation.

On a compact manifold without boundary with zero Gaussian curvature $K = 0$, the maximum principle applies to harmonic functions: Φ achieves its maximum and minimum on the boundary. Since T^3 has no boundary, Φ must be constant everywhere. The only harmonic functions on T^3 are constants, so $\nabla\Phi = \mathbf{0}$: the only divergence-free *potential* flow on T^3 is trivially zero.

This does not mean that all flows are zero. The flat metric of T^3 permits *uniform flows* $\mathbf{v} = \mathbf{v}_0 = \text{const}$ — constant vector fields that are clearly divergence-free ($\nabla\cdot\mathbf{v}_0 = 0$) and consistent with the periodic boundary conditions. These are not derivable from a scalar potential (since $\nabla\Phi = \mathbf{v}_0 \neq \mathbf{0}$ would require Φ to grow linearly, contradicting periodicity). The uniform flows are the non-trivial laminar flows on T^3: constant-speed propagation along straight geodesics,

with no pressure gradient, no differential acceleration, and no mechanism for turbulence to develop.

The key consequence for information: on T^3, *any* divergence-free flow is a uniform flow. There is no other option. The flat geometry and the absence of a boundary together force constant-velocity geodesic motion as the unique admissible flow type.

6.3 Why Curvature Would Destroy Laminarity

The $K = 0$ condition is not merely convenient — it is necessary for laminarity. On a curved manifold (such as a sphere S^3 or a hyperbolic space), geodesics converge or diverge, creating regions of higher and lower information density. A uniform initial flow becomes non-uniform after propagating through a curved region, creating differential arrival times and potential interference between states propagating along different geodesics. On T^3, all geodesics are straight lines (with periodic identification), and they never converge or diverge. The laminarity is a consequence of the flatness: $K = 0$ is the condition for uniform geodesic flow to be possible, and the compactness is the condition for it to be necessary.

6.4 Quantised Circulation and Flow Locking

The superfluid structure adds a quantisation constraint on top of the laminar flow:

$$\Gamma_i = \oint_{C_i} \mathbf{v} \cdot d\mathbf{l} = n_i \frac{h}{m}, \qquad n_i \in \mathbb{Z}. \tag{6}$$

The circulation around each generating circle is an integer multiple of h/m, locking the flow to the discrete winding structure. The flow velocity and the winding address are the same object from two perspectives: the circulation $(\Gamma_1, \Gamma_2, \Gamma_3)$ encodes the same information as the winding vector (w_1, w_2, w_3). The laminar flow is not an independent property layered on top of the winding structure — it *is* the winding structure, expressed in the language of circulation.

6.5 Information Velocity and Latency

Laminar flow on T^3 propagates at speed c (the signal velocity in the medium). On the torus with side length L, the maximum distance between any two points — measured along geodesics with periodic identification — is $L/2$ per direction, giving a maximum three-dimensional separation of $L\sqrt{3}/2$. The maximum propagation time from any state to any other state is therefore

$$t_{\max} = \frac{L\sqrt{3}}{2c} < \sqrt{3}\, t_q. \tag{7}$$

The information latency (maximum propagation delay) is bounded by $\sqrt{3}$ clock periods, independent of the physical size of the system. Two states in the same vocabulary can exchange information within two clock periods. This latency bound is topological: it follows from the geometry of T^3, not from the specific dynamics of the medium.

6.6 The Connection to Noise Rejection

Laminar flow has a direct consequence for the noise rejection mechanism studied in the next section. A noisy state — one with $|w_i| \geq 2$ for some component — cannot dissipate its excess winding gradually into the background flow, because the background flow has no turbulent channel to absorb it. The uniform flow simply carries the noisy state intact. The only available dissipation process is the discrete Landau decay: the state must split into unit-winding states. Laminar flow thus *enforces* the Landau filter by eliminating the alternative: gradual winding dissipation into a turbulent background is impossible on T^3.

7 Requirement 5: Noise Rejection and Forbidden Information Shells

7.1 Classes of Noise

In the context of this information system, noise is any state not in Ω_{stable}. Three distinct classes arise:

1. *High-winding states*: $|\mathbf{w}|^2 \geq 4$, violating the Landau criterion. These states *exist* as Laplacian eigenstates of T^3 but are energetically unstable.
2. *Geometrically absent states*: energy shells $|\mathbf{n}|^2 \in \{7, 15, 23, \ldots\}$ for which no Laplacian eigenstate exists on T^3.
3. *Superposition states*: linear combinations of stable states without a definite winding address, filtered by the discreteness of $\mathbb{Z}^3$.

The first two classes are handled by the two filters described below. The third class is handled automatically by the integer structure of winding numbers: a superposition of winding states is not itself a winding state; it decays via interference into a definite winding configuration on the timescale set by ε_0.

7.2 The Landau Filter: Energetic Self-Correction

Any state with $|\mathbf{w}|^2 > 3$ is energetically unstable (Theorem 1) and spontaneously decays within relaxation time $\tau \approx t_q$. The Landau filter is *dynamical*: it acts on states that are present in the system but energetically disfavoured. As established in Section 6, laminar flow ensures that there is no turbulent channel to absorb the excess winding; the state must decay by splitting.

The energy barrier for the decay of a state with $|\mathbf{w}|^2 = 4$ (the first unstable shell) is

$$\Delta E = E_{\text{original}} - E_{\text{split}} = 4\varepsilon_0 - (1 + \text{smaller pieces})\varepsilon_0 \geq 2\varepsilon_0. \tag{8}$$

The energy is released as phonons or other collective excitations of the superfluid. The decay products are always members of Ω_{stable}, so the system self-corrects without any external control.

7.3 The Legendre Filter: Geometric Non-Existence

The second filter operates at a deeper level. By Legendre's three-square theorem (Appendix B), a positive integer s is expressible as $n_1^2 + n_2^2 + n_3^2$ for integers (n_1, n_2, n_3) if and only if $s \notin \{4^a(8b+7) : a, b \geq 0\}$. The first forbidden values are $s \in \{7, 15, 23, 28, 31, \ldots\}$.

The Laplacian on T^3 has eigenvalues $(2\pi/L)^2|\mathbf{n}|^2$, and an eigenstate at shell $|\mathbf{n}|^2 = s$ exists if and only if s can be written as a sum of three integer squares. For Legendre-forbidden s, no such triple exists: the eigenstate has no carrier. This is not energetic suppression — it is structural absence. A noise signal at frequency $c\sqrt{s}/L$ for forbidden s has *no mode to occupy.* It cannot propagate in T^3 regardless of amplitude, source, or energy.

7.4 The Three-Layer Protection Zone

The gap between the stable vocabulary and the first geometrically absent shell creates a three-layer protection zone above the vocabulary range $|\mathbf{w}|^2 \leq 3$:

Shell $\|\mathbf{n}\|^2$	Exists?	Status	Protection mechanism
1,2,3	Yes	**Stable vocabulary**	Landau criterion satisfied
4,5,6	Yes	Landau unstable	Energetic decay in $\sim t_q$
7	**No**	Legendre absent	No carrier exists
8–14	Yes	Landau unstable	Energetic decay
15	**No**	Legendre absent	No carrier exists
$\vdots$	$\vdots$	$\vdots$	$\vdots$

The two filters are complementary, not redundant. The Landau filter handles shells 4–6 (and 8–14, etc.): these exist as eigenstates but decay. The Legendre filter handles shell 7 (and 15, 23, etc.): these do not exist at all. A noise signal trying to "rest" at shell 7 after failing to be absorbed by shell 4–6 finds that shell 7 does not exist as a mode. The combined effect is that after one clock period, any signal not in Ω_{stable} has either decayed (Landau) or had nowhere to exist (Legendre). The system self-restores without external intervention.

7.5 Quantitative Noise Suppression

The energy required to create any noise state above Ω_{stable} is at least $\Delta E_{\text{noise}} = (4-3)\varepsilon_0 = \varepsilon_0$. At temperature T, the thermal probability of spontaneously creating such a state is suppressed by the Boltzmann factor $e^{-\varepsilon_0/k_B T}$. For the system to be in the stable information regime, we require $k_B T \ll \varepsilon_0$, which translates to $T \ll T_c$ (well below the superfluid transition temperature). This condition defines the operating regime of the information system.

8 The Complete Closed Information System

8.1 Why Five Requirements Are Sufficient for Closure

Sections 3–7 established that each of the five requirements is satisfied. But satisfying five requirements separately does not automatically constitute a *closed* system. We now argue that the five together are jointly sufficient.

A closed information system must satisfy three meta-conditions:

1. No information escapes: guaranteed by $\partial T^3 = \varnothing$ and conservation of winding numbers.
2. No information is internally lost: guaranteed by the seven conservation laws and the closure of Ω_{stable} under transitions.
3. Perturbations are expelled: guaranteed by the Landau-Legendre double filter and the laminar flow structure that prevents diffusion of noise.

Each meta-condition is met. The system does not merely *satisfy* five requirements — the five requirements together *constitute* the closure.

8.2 Assembly of the Five Requirements

Requirement	Source	Result
Stable states	Landau + $\pi_1(T^3) = \mathbb{Z}^3$	$26 = 3^3 - 1$ states, 3 families
Internal clock	Cavity eigenspectrum	$f_1 = c/L$; harmonics nf_1
Conservation	Noether + topological winding	7 conserved quantities
Laminar flow	$K = 0$ + Laplace, compact T^3	Unique uniform geodesic flow
Noise rejection	Landau + Legendre	Self-restoring in $\sim t_q$

8.3 Information Capacity

The maximum information capacity of the stable vocabulary is

$$H_{\max} = \log_2(26) \approx 4.70 \text{ bits per address}, \tag{9}$$

assuming uniform occupancy (maximum entropy). The $3^3 = 27$ lattice points of $\{-1, 0, 1\}^3$ divide as one vacuum + 26 vocabulary states; the 26 states require $\lceil \log_2 26 \rceil = 5$ bits to address, corresponding to $26/32 \approx 81\%$ density in 5-bit address space.

For context, this vocabulary is larger than the 20-letter amino acid alphabet (20 states < 26), richer than a 4-bit system (16 states), and more compact than a 5-bit system (32 states). The T^3 information system sits at a natural combinatorial sweet spot: the maximal vocabulary achievable with three binary winding directions under the Landau stability constraint.

8.4 The Three-Channel Architecture

The three families constitute a natural hierarchy of channel utilisation:

- **Face states** (6): single-channel encoding. One winding direction active. Maximally isolated from cross-channel coupling. These states have the lowest energy (ε_0) and the longest lifetime.
- **Edge states** (12): dual-channel encoding. Two winding directions active. Intermediate energy ($2\varepsilon_0$). Information in one channel is partially recoverable from the second channel: if one winding number is perturbed to zero, the remaining channel still encodes partial information.
- **Corner states** (8): triple-channel encoding. All three winding directions active. Highest energy ($3\varepsilon_0$). Maximum coupling to all three conservation laws. These states are the most "entangled" with the full topology of T^3.

By Theorem 2, no symmetry of the system interchanges families: they are structurally inequivalent, not merely conventionally labelled differently.

8.5 State Transition Rules

Since winding numbers compose additively in $\mathbb{Z}^3$, the allowed transitions $\mathbf{w} \to \mathbf{w}'$ must satisfy $\mathbf{w}' = \mathbf{w} + \Delta\mathbf{w}$ where $\Delta\mathbf{w} \in \Omega_{\text{stable}} \cup \{\mathbf{0}\}$. This means:

- A state can transition to any neighbour in $\mathbb{Z}^3$ that is itself stable — a ± 1 change in any component.
- Opposite states $\mathbf{w}$ and $-\mathbf{w}$ can annihilate to the vacuum.
- The vacuum can produce any pair $(\mathbf{w}, -\mathbf{w})$ with energy input $2|\mathbf{w}|^2\varepsilon_0$.
- The vocabulary Ω_{stable} is closed: no transition within it leaves it.

The transition graph on Ω_{stable} is the Cayley graph of $\{-1, 0, 1\}^3$ under the action of Ω_{stable}: a regular structure with high connectivity that supports arbitrary information routing within the vocabulary.

8.6 Spin Structures and Signal Polarity

T^3 admits exactly $2^3 = 8$ distinct spin structures, classified by $H^1(T^3, \mathbb{Z}_2) \cong \mathbb{Z}_2^3$ [6]. Each assigns a boundary condition $\rho_i \in \{+1, -1\}$ around each generating circle: periodic ($\rho_i = +1$) or anti-periodic ($\rho_i = -1$). The choice of spin structure determines whether the spinor field has integer or half-integer momentum eigenvalues in each direction. The eight spin structures provide eight distinct signal polarity classes, adding a binary label in each of three dimensions on top of the winding address. States with the same winding vector but different spin structure are physically distinguishable by their momentum eigenvalue spectrum.

8.7 The Vocabulary as a Finite-State Automaton

The 26-state vocabulary together with the transition rules defines a finite-state automaton (FSA) with 26 states and transitions determined by the $\mathbb{Z}^3$ group structure. This FSA is symmetric (the transition relation is the same as its inverse, since $\Delta\mathbf{w}$ and $-\Delta\mathbf{w}$ are both stable), regular (each state has the same number of reachable neighbours), and connected (any state can be reached from any other via a sequence of transitions).

The question of whether this FSA is computationally universal is open. A sufficient condition for universality would be the ability to implement arbitrary reversible Boolean operations on the winding addresses. The closure of Ω_{stable} under transitions and the richness of the $\mathbb{Z}^3$ structure suggest that the system can implement a significant class of reversible computations, but a complete characterisation is left for future work.

9 Boundaries of the Framework

9.1 Formal Definition of Closure

The word "closed" has been used throughout this paper. It has a precise meaning here: a closed information system is one in which (1) no information escapes, (2) no information enters uninvited, and (3) all internal information is conserved. The T^3 system satisfies all three:

1. No escape: $\partial T^3 = \varnothing$ means there is no boundary through which information could exit.
2. No injection: the same condition prevents uninvited injection; the boundary is absent in both directions.
3. Internal conservation: seven conservation laws ensure that neither the winding address nor the dynamical state can spontaneously vanish.

These three conditions are not independent: (1) and (2) follow from the same topological property ($\partial T^3 = \varnothing$), and (3) follows from the structure of T^3 as a Lie group quotient with $\pi_1 = \mathbb{Z}^3$. Closure is a single package delivered by the topology.

9.2 What Breaking Closure Would Require

The closure is robust: it cannot be broken by continuous perturbations of the field configuration, only by changes to the topology itself. To break the closure, one would need to:

- *Add curvature*: $K \neq 0$ would destroy laminarity by creating geodesic focusing and turbulence. This requires deforming T^3 into a curved manifold — a topological change, not a continuous field perturbation.
- *Reduce dimension*: $b_1 < 3$ would reduce the vocabulary to $3^{b_1} - 1$ states, eliminating the three-family structure. This requires changing the manifold, not just its fields.

- *Open the boundary*: $\partial T^3 \neq \varnothing$ would allow information leakage and injection. This also requires changing the manifold.
- *Remove the superfluid*: without the $U(1)$ order parameter, the winding numbers have no physical realisation. The system loses its information carrier.

The stability of the closure against continuous perturbations is itself a topological fact: the manifold must be changed, not merely perturbed.

9.3 External Parameters

The structural properties are independent of four external scale parameters:

Parameter	Role	Dependence
Scale L	Clock period, energy unit	$f_1 = c/L$; $\varepsilon_0 \propto \ln(L/\xi)$
Signal speed c	Information velocity	$f_1 = c/L$; $v_{\max} = c$
Superfluid density ρ_s	Vortex energy scale	$\varepsilon_0 \propto \rho_s$
Coherence length ξ	Vortex core size	$\varepsilon_0 \propto \ln(L/\xi)$

Changing any of these parameters rescales the clock frequency, the energy barrier, and the relaxation times — but leaves the 26-state vocabulary, three-family structure, seven conservation laws, and noise filter boundaries unchanged. Two T^3 systems with different (L, c, ρ_s, ξ) are structurally identical: same vocabulary, same addressing, same conservation laws. The structure is topological; the parameters are material.

9.4 What the Framework Does Not Address

This paper characterises T^3 as an information environment. It does not address:

- **External dynamics**: cosmological expansion, dark energy, and gravity are features of an open universe, excluded by definition from a closed T^3 system.
- **Internal dynamics**: specific transition probabilities, interaction cross-sections, and decoherence times depend on the Lagrangian governing the superfluid, not on the topology of T^3.
- **The origin of L**: why the system has a particular scale is a dynamical question, not a topological one.
- **External coupling**: how information enters or exits from an environment requires specifying an interface. The topology of T^3 alone does not determine this.

These are genuine boundaries, not omissions. The framework determines the structure; the dynamics and the interface are separate questions.

10 Summary

Property	Value / Structure	Source
Stable vocabulary	$3^3 - 1 = 26$ states	Landau stability on T^3
Family structure	3 families $(6 + 12 + 8)$	$b_1(T^3) = 3$
Fourth family	Forbidden	Landau: $\|\mathbf{w}\|^2 = 4$ decays
$\mathbb{Z}_3$ charge partition	$8 + 9 + 9$	$T^3/\mathbb{Z}_3$ holonomy
Symmetry group	O_h, order 48	Cubic lattice symmetry
Root system (face+edge)	$B_3 = \mathfrak{so}(7)$, 18 roots	Cartan matrix
Fundamental clock	$f_1 = c/L = c/(2\pi R)$	Cavity eigenspectrum
Clock-state correspondence	$f_i \propto \sqrt{E_i}$	Same integer lattice
Time quantum	$t_q = L/c$	Inverse clock frequency
Continuous conservation	4 (energy + 3 momenta)	Noether + translations
Topological conservation	3 winding numbers	$\pi_1(T^3) = \mathbb{Z}^3$
Total conservation	7	Noether + topological
Error gap	ε_0	Topological winding threshold
Flow type	Laminar, uniform geodesic	$K = 0$ + Laplace
Information latency	$\leq \sqrt{3}\, t_q$	T^3 geometry
Landau filter	Decays within t_q	$E(n\mathbf{w}) = n^2 E(\mathbf{w})$
Legendre filter	Forbidden shells $4^a(8b + 7)$	Three-square theorem
First forbidden shell	$\|\mathbf{n}\|^2 = 7$	Legendre
Addressing capacity	$\log_2(26) \approx 4.70$ bits	26-state vocabulary
Addressing bits	$\lceil \log_2 26 \rceil = 5$	
Spin structures	$8 = 2^3$	$H^1(T^3, \mathbb{Z}_2) \cong \mathbb{Z}_2^3$
Transition structure	FSA on Ω_{stable}	$\mathbb{Z}^3$ group law
External parameters needed	L, c, ρ_s, ξ (scale only)	Topology-independent

11 Conclusions

The flat torus $T^3 = \mathbb{R}^3/L\mathbb{Z}^3$ provides, by mathematical necessity, all five requirements for a closed information-processing environment. Each requirement is met by a specific mathematical object that the topology provides, and the five together are jointly sufficient for closure.

Stability is provided by the Landau criterion on the $U(1)$ superfluid order parameter: the n^2 energy scaling restricts stable states to components in $\{-1, 0, 1\}$, producing exactly $3^3 - 1 = 26$ states in three structurally non-isomorphic families. The stability is absolute within the energy regime $k_B T \ll \varepsilon_0$: there is no noise floor to manage, only a topological gap to maintain.

Clocking is provided by the cavity resonance spectrum of T^3: the fundamental frequency $f_1 = c/L$ is uniquely determined by the manifold scale and signal speed, with the clock and the vocabulary both indexed by the same integer lattice $\mathbb{Z}^3$. The clock-state correspondence is intrinsic to the geometry.

Conservation is provided by seven independent laws from two different sources: four continuous Noether charges from the translational symmetries, and three topologically protected winding numbers from $\pi_1(T^3) = \mathbb{Z}^3$. The topological charges have the further property of an error gap: perturbations smaller than ε_0 cannot alter a winding address.

Laminarity is the unique consequence of the Laplace equation on a compact flat manifold: the only admissible flow is uniform constant-velocity geodesic motion, with an information latency bounded by $\sqrt{3}$ clock periods. Crucially, the laminarity enforces the Landau filter: without a turbulent background to absorb noisy states, those states must decay by splitting.

Noise rejection operates at two complementary levels. The Landau filter dynamically decays

states above the vocabulary range within one clock period. The Legendre filter geometrically forbids states at certain shells from ever having a carrier, creating an absolute noise gap just above the Landau range.

The results that are not obvious from the starting assumptions:

- The $3^3 - 1$ formula: the vocabulary is exactly the non-zero elements of $\{-1, 0, 1\}^3$. No larger stable vocabulary is possible under the Landau criterion on T^3.
- The non-isomorphism is unconditional: the three $\mathbb{Z}_3$-module structures are provably distinct, forcing a mandatory coupling hierarchy on any interaction that distinguishes the families.
- The clock-state correspondence: the clock spectrum and the energy spectrum are indexed by the same integer lattice, not by coincidence but by construction.
- The Legendre gap protects against re-entry: the first forbidden shell $|\mathbf{n}|^2 = 7$ appears just above the Landau decay range $|\mathbf{n}|^2 \in \{4, 5, 6\}$, ensuring that a decaying noise state cannot "park" at an intermediate shell.
- The closure is topological: it cannot be broken by continuous field perturbations. Only a change of manifold can break it.

The mathematical properties of the T^3 closed information system are entirely determined by the choice of manifold. The external parameters L, c, ρ_s, ξ set the physical scale; the topology sets the structure.

Appendix A Verification of the B_3 Root System

We verify that the 18 face-and-edge vortex states form $B_3 \cong \mathfrak{so}(7)$. Let $\mathcal{R} = \{\text{face states}\} \cup \{\text{edge states}\}$, with $|\mathcal{R}| = 18$.

1. **Closure under negation.** For each $\mathbf{w} \in \mathcal{R}$, $-\mathbf{w} \in \mathcal{R}$. ✓
2. **No non-± 1 multiples.** For any $\mathbf{w} \in \mathcal{R}$, $2\mathbf{w} \notin \mathcal{R}$. ✓
3. **Integer Cartan entries.** For all $\alpha, \beta \in \mathcal{R}$, $\langle \alpha, \beta \rangle = 2(\alpha \cdot \beta)/(\beta \cdot \beta) \in \mathbb{Z}$. (Verified for all $18 \times 17 = 306$ ordered pairs.) ✓
4. **Reflection closure.** For all $\alpha, \beta \in \mathcal{R}$, $\beta - \langle \alpha, \beta \rangle \alpha \in \mathcal{R}$. (Verified for all pairs.) ✓

The simple roots are $\alpha_1 = (1, -1, 0)$, $\alpha_2 = (0, 1, -1)$, $\alpha_3 = (0, 0, 1)$, with Cartan matrix

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -1 & 2 \end{pmatrix}. \tag{10}$$

The Weyl group has order $3! \times 2^3 = 48 = |O_h|$, confirming that O_h is the full Weyl symmetry of the root system.

Appendix B Legendre's Three-Square Theorem

Theorem (Legendre, 1798 [7]). A positive integer n is expressible as $a^2 + b^2 + c^2$ for $a, b, c \in \mathbb{Z}$ if and only if n is not of the form $4^a(8m + 7)$ for non-negative integers a, m.

The proof uses the theory of ternary quadratic forms [8]. The first forbidden values are $7, 15, 23, 28, 31, 39, 47, 55, 60, 63, 71, \ldots$

The T^3 Laplacian has no eigenstate at shell $|\mathbf{n}|^2 = s$ when s is forbidden: no $\mathbf{n} \in \mathbb{Z}^3$ satisfies $|\mathbf{n}|^2 = s$. The shells are geometrically absent, not merely energetically disfavoured. Information at these frequencies cannot propagate on T^3 regardless of source or amplitude.

The smallest forbidden shell $s = 7$ is significant: it sits just above the Landau decay range $|\mathbf{n}|^2 \in \{4, 5, 6\}$. A noise signal decaying from the Landau range cannot find a carrier at $s = 7$; it must continue decaying until it reaches the stable vocabulary at $|\mathbf{n}|^2 \leq 3$. This enforces the two-step descent: first Landau decay moves the signal below $|\mathbf{n}|^2 = 4$; then a re-entry attempt at $|\mathbf{n}|^2 = 7$ finds nothing. The combined effect is that any signal outside Ω_{stable} descends to the vocabulary without intermediate trapping.

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